

# Omar Alghannam

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## EDUCATION

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**Zewail City of Science and Technology** | Sep 2024 – May 2028 | GPA: 2.8

*B.Sc. in Data Science and Artificial Intelligence — 6th of October City, Giza*

Relevant Coursework: Data Structures, Probability, Python Programming, Linear Algebra, Database, Machine Learning, Data Mining, Algorithms

## EXPERIENCE

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**AI/ML Intern — Advanced AI & Data Analytics** | e& Egypt · Hybrid

*Aug 2026 – Sep 2026*

- Designed and built an agentic, bilingual (English/Arabic) RAG chatbot using LangGraph, ChromaDB, and Azure OpenAI — featuring a 7-node self-correcting pipeline with query expansion, intent routing, two-stage retrieval (dense semantic search + cross-encoder reranking), and automated answer validation with self-healing retries.
- Validated system reliability through RAGAS evaluation (faithfulness, answer relevancy), LLM-as-judge scoring against a 50-case ground-truth test suite, and automated red-teaming via Garak to verify prompt injection and jailbreak resistance.
- Co-engineered an end-to-end B2B Churn Intelligence Platform utilizing Python, FastAPI, XGBoost, and SHAP, delivering real-time, explainable risk scoring alongside executive dashboards and squad management tooling.
- Architected a React frontline retention portal featuring role-based access control (RBAC), dynamic call queues, live quota tracking, and an in-call GenAI copilot generating context-aware retention talking points.

**Microsoft Data Engineer Intern** | Digital Egypt Pioneers Initiative (DEPI) · Remote

*Dec 2025 – Jul 2026*

- Trained in building and managing modern data systems supporting analytics, AI, and business intelligence solutions.
- Developed practical skills in database management, data pipelines, big data processing, and cloud-based data engineering using Microsoft Azure technologies (Azure Data Factory, Azure Synapse Analytics, Databricks).
- Focused on scalable data architectures, deployment practices, and AI-driven data workflows through hands-on projects aligned with industry demands.

**AI/ML Engineer – LLM Fine-Tuning** | Freelance Contract (Upwork) · Remote

*Jun 2026*

- Engineered an end-to-end fine-tuning pipeline for Gemma-4 (27B) using PyTorch, HuggingFace PEFT, and LoRA to adapt large-scale models for specialized customer support automation.
- Implemented custom loss-masking logic for reasoning models, enabling dynamic supervision of internal reasoning traces (<think> blocks) versus final generative outputs to optimize model logic and factual accuracy.
- Optimized models for high-performance inference by merging LoRA weights and utilizing NVIDIA modelopt for advanced NVFP4 calibration, enabling ultra-fast deployment in vLLM.

## PROJECTS

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**UK Road Accident Analytics Platform (DEPI Graduation Project)**

*Airflow, FastAPI, React, PostgreSQL, Docker, Azure* July 2026

- Engineered an automated ETL pipeline using Apache Airflow to ingest, clean, and process over 500,000 historical accident records into a PostgreSQL data warehouse.
- Built a high-performance FastAPI backend to serve real-time analytics and Machine Learning severity predictions to a live, interactive React dashboard.
- Containerized the entire microservices architecture using Docker and deployed the full-stack system to Microsoft Azure for scalable, cloud-based production.

**CineMatch AI – Recommendation System**

*FastAPI, PyTorch, Supabase* March 2026

- Engineered a personalized AI movie platform using TMDB API to construct dynamic user profiles based on real-time interactions.
- Optimized candidate retrieval using TF-IDF and SBERT embeddings; built hybrid ranking pipeline with SVD collaborative filtering and PyTorch Neural Network via 64-dimensional embeddings.

- Deployed scalable FastAPI/Supabase backend with automated model retraining, verified through NDCG@10 and Precision@10 metrics.

### **Football Search Engine**

*Python, Streamlit, DuckDB, Ollama, SerpAPI* 2025

- Built a full-featured football search engine with multiple ranking algorithms: TF-IDF, Semantic (nomic embeddings), Word2Vec Skip-gram, and a weighted Hybrid mode.
- Implemented query expansion via LLM (Ollama llama3.2), Word2Vec, and co-occurrence matrix; word representations include One-Hot, CBOW, Skip-gram, and GloVe-style SVD.
- Integrated SerpAPI for dynamic corpus growth, DuckDB for persistent storage, and IR evaluation metrics (Precision, Recall, F1, MAP, nDCG).

### **Mowasla – Uber-like Ride-Hailing System**

*C#, SQL* Dec 2025

- Developed backend in C# with SQL database modeling Passenger, Driver, Vehicle, Ride, and Payment entities.
- Built passenger features (ride requests, fare estimation) and driver features (accept/reject, earnings dashboard).
- Designed admin panel for user management, ride monitoring, and system statistics generation.

### **Premier League Analytics Dashboard**

*AmCharts 5, Data Visualization* Dec 2025

- Built interactive dashboard analyzing the 2023-24 Premier League season with advanced visualizations (bar, scatter, donut, area charts) using AmCharts 5.
- Visualized KPIs (goals, assists, squad value, wage efficiency, revenue) with dynamic filters; delivered actionable analytics on player performance and club ROI.

## **TECHNICAL SKILLS**

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**Languages:** Python, C++/C, C#, SQL, JavaScript, TypeScript, HTML/CSS

**Backend:** FastAPI, Flask, REST APIs, Node.js, System Design

**ML/AI:** PyTorch, Scikit-learn, TensorFlow, NLP, LLM Fine-tuning (LoRA/PEFT), RAG, LangChain, Recommendation Systems

**Data:** Pandas, NumPy, Data Analysis, Data Visualization, Azure Data Factory, Azure Synapse Analytics, Databricks

**Tools:** Git, Firebase, Supabase, DuckDB, n8n, Streamlit, Playwright, Selenium, BeautifulSoup, vLLM, HuggingFace

**Frontend:** React, Next.js, Tailwind CSS

## **ACHIEVEMENTS**

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- Certificate of Participation – 20th Undergraduate Research Forum (UGRF), Nile University, Egypt (Aug 2025) — Project: Cleaning System Robot